

Design and behavior of 160 m-tall post-tensioned precast concrete-steel hybrid wind turbine tower

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Abstract. Prefabricated hybrid wind turbine towers (WTTs) are promising due to height increase. This study proposes the use of ultra-high performance concrete (UHPC) to develop a new type of WTT without the need to use reinforcement. It is demonstrated that the UHPC WTT structure without reinforcing bars could achieve performance similar to that of reinforced concrete WTTs. To simplify the design of WTT, a design approach for the calculation of stresses at the horizontal joints of a WTT is proposed. The stress distribution near the region of the horizontal joint of the WTT structure under normal operating conditions and different load actions is studied using the proposed approach, which is validated by the finite element method. A further parametric study shows that the degree of prestressing and the bending moment both significantly affect the principal stress. The shear-to-torsion ratio also shows a significant influence on the principal tensile stress.

Keywords: design; finite element; hybrid; post-tensioned; precast concrete; stress; UHPC; wind turbine towers

1. Introduction

Wind energy is a form of renewable energy experiencing rapid growth globally. Wind at higher elevations is faster and more sustainable, which has resulted in the trend whereby the height of wind turbine tower (WTT) has increased from 80 m to more than 100 m. This has encouraged the development of new types of WTTs and new design approaches. In recent years, hybrid WTT-which consists of a lower part made of prestressed concrete segments and an upper part made of a steel segment-has attracted growing attention (Fig. 1). The combination of concrete and steel in a WTT can improve both the static and dynamic performances of the WTT and reduce the total cost significantly. Further, the concrete segments can be a prefabricated assembly, which reduces the formwork.

For hybrid WTTs under normal operating conditions, the tensile stress at the concrete part or near the door opening may exceed the tensile strength of concrete and cause tensile failure of concrete, as normal concrete has low tensile strength. This highly compromises the durability and safety of a WTT structure. Ultra-high performance concrete (UHPC) exhibits superior mechanical properties with a



Fig. 1 Post-tensioned hybrid WTT consisting of precast concrete and steel segments

tensile strength ranging from 7 MPa to 15 MPa, which is much greater than that of normal concrete. This provides a new option for solving the aforementioned issue. UHPC is also considered to be an alternative material to traditional steel WTTs. Schmitz and Peggarr and Srithran both proposed designs of UHPC WTTs and experimentally validated a 100 m tall WTT. Yang also studied the behaviors of the UHPC towers with different wall thicknesses under rated wind speed, limit wind speed, and stormwind speed using the finite element (FE) method, and they obtained the principal stress along the height of the tower. However, the utilization

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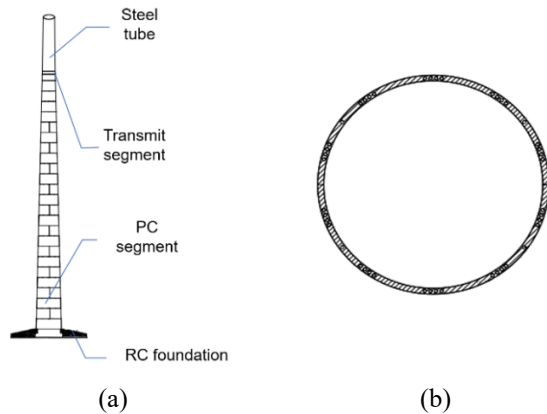


Fig. 2 (a) Sketch of a typical post-tensioned hybrid WTT and (b) a typical cross-section of a WTT

of UHPC in WTTs and relevant studies in this field still remain limited.

Further, as a relatively new type of structure, recent studies on WTTs have mainly focused on the design approach and stress analysis. FE and analytical methods have both been applied on a WTT to understand its behavior under different loading conditions. Different types of WTT structures, such as tubular steel towers, concrete towers, prestressed concrete towers, etc., have been widely studied as well. For example, Quilligan conducted a fragility analysis on a WTT with a height ranging from 88 m to 120 m. In another study, Tao investigated the static strength and fatigue strength of the joints of ring units to optimize the design of WTTs. Li found that the fatigue failure of the joint of a WTT was caused by the local high stress after cyclic disturbance loads. Zhang studied the dynamic behavior of a 1.5 MW three-bladed WTT under full load conditions and found that using concrete for a WTT can effectively reduce the propagation of cracks at the joint of the tower and ultimately improve its service performance. Bazeos et al. conducted the static and dynamic analyses of a WTT under three working conditions and found that the cross-sectional stress of the tower decreased with an increase in the tower height, and that the stress of the tower typically changed suddenly at the locations experiencing changes in wall thickness and diameter. Most of the analyses in the current studies were conducted through FE modeling.

Based on these studies, it was found that, in the prefabricated WTT, the stress may increase significantly at the joints with an increasing number of ring units. This has led to interest in further investigating the stress distribution of a WTT at joints and to optimize the design approach. Alvarez-Anton proposed the optimization of hybrid WTTs based on Building Information Modeling (BIM) and prefabrication techniques for tall WTTs. In another study, Stanley et al. optimized the turbine design in wind farms with multiple hub heights. Pons also optimized the design of hybrid WTTs with a hub height of 120 m to reduce the cost of the tower. However, there is still a lack of design approaches focusing on the joints of segments, which are critical parts of WTTs. To understand the behavior of a WTT and its structural design, it is greatly important to



Fig. 3 Typical precast concrete segments for a WTT

have a simple calculation method for the stress distribution of the joints of segments.

In this study, UHPC is first proposed to replace the high-strength concrete and reinforcing bars in a WTT. The feasibility of a UHPC WTT is then validated by comparing it with the reinforced high-strength concrete WTT. A design approach on the joints of segments is also proposed to determine the stress distribution at the cross-sectional joints of a WTT.

2. Analysis and behavior of UHPC WTT

2.1 Finite element modeling

(1) WTT structure and materials

As shown in Fig. 2, a typical prefabricated hybrid WTT structure is generally composed of an upper steel tubular segment, a steel transition segment, precast concrete segments, and a foundation. A typical hybrid WTT structure from the authors' previous work is used in this study. The height of the bottom concrete part of the WTT is 110.5 m. The diameter of the structure is 8 m at the bottom of the tower, and it is gradually reduced from the bottom to the top. The wall thickness of the tower is between 220 mm and 460 mm. The tower consists of 28 sections with a height of approximately 4 m for each section (Fig. 3). The sections are numbered from the bottom to the top and denoted by M and the section number.

To investigate the feasibility of using UHPC in the WTT, two types of structures with different types of concrete are considered for comparison and FE models are established accordingly by using the commercially available ABAQUS software. The two models have the same design conditions of geometry and are both made of concrete segments. In the first model, which is denoted as WTT-H160/C80, the concrete with compressive strength of 80 MPa (C80) is used with steel reinforcement. In the other model, denoted as WTT-H160/UHPC, the UHPC with compressive strength of 120 MPa (UHPC120) is used but the steel reinforcement-except for the prestressing steel tendons-has been removed. In both models, the concrete is reinforced or post-tensioned by HRB400 ordinary hot-rolled steel bars and/or high-strength prestressing steel tendons with six 7-wire prestressing steel strands per tendon. Table 1 summarizes the properties of the materials used in this

Table 1 Material properties

Material	Modulus of elasticity (MPa)	Poisson's ratio	Mass density (kg/m ³)	Yield strength (MPa)	Expansion coefficient (°C ⁻¹)
C80	3.8×10^4	0.2	2400	-	-
UHC120	4.0×10^4	0.3	2400	-	-
HRB400	2.0×10^5	0.3	7850	360	1.2×10^{-5}
Prestressed steel strand	1.95×10^5	0.3	7850	3.6×10^8	1.2×10^{-5}

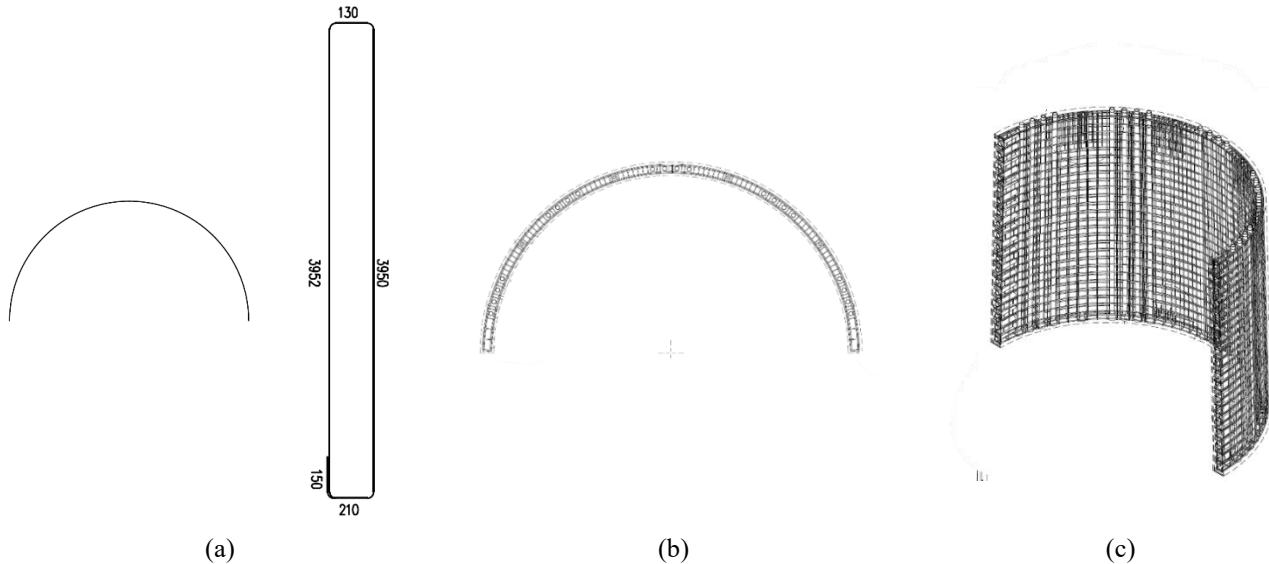


Fig. 4 Reinforcement details of WTT-H160/C80: (a) circumferential and longitudinal reinforcing bars; (b) top view of reinforcement in a concrete segment; and (c) total reinforcement detail in a concrete segment (Unit: mm)

study. All tower materials are assumed to be elastic under normal operating conditions.

(2) Interaction

Ordinary steel bars are divided into circumferential and longitudinal reinforcing bars. The WTT-H160/C80 has steel reinforcement. Fig. 4(a) shows circumferential and longitudinal reinforcing bars used in the model. The total steel reinforcement details in a concrete segment are indicated in Figs. 4(b) and 4(c). Table 2 shows the circumferential and longitudinal reinforcement details of 28 concrete segments. The higher the segment, the lower the amount of longitudinal reinforcement is required. The circumferential reinforcement is constant. In this study, the “embedded technique” of ABAQUS is used to simulate the synergistic relationship between steel bars and concrete and the connection between prestressing reinforcement and concrete in a WTT structure. Here, it is assumed that there is no relative displacement between the steel bars and the concrete elements.

In practice, structural glue is typically applied at the joint section of a WTT and the tensile strength of the structural glue at the joint is greater than those of the other parts. Therefore, the WTT structure can be simulated as a whole model. In the FE model, the whole structure is divided into three parts. The middle part from M2 to M27 is established as an intact segment, while the bottom and top parts are established separately and connected with the middle part by a “tie” in ABAQUS.

(3) Boundary conditions

It is assumed that the WTT and foundation are completely fixed. To avoid the local damage caused by stress concentration at the top of the tower, a steel gasket with a thickness of 10 cm is set at the top of WTT in practice. This was simulated as a rigid constraint in ABAQUS. In addition, the interaction between the steel gasket and M28 is also modeled by a “tie”.

(4) Element type and mesh size

C80 and UHPC are simulated by the 8-node hexahedral linear reduced integral elements (C3D8R). The three-dimensional two-node linear displacement truss elements (T3D2) are used to simulate the reinforcement.

Since the diameter of the structure is gradually reduced from the bottom to the top, the local seed was applied in the model to control the mesh and ensure the coordination of meshes between segments at the joints. In the thickness direction, the wall of the WTT was meshed into three elements. Further, the mesh was refined locally at the bottom near the door opening.

(5) Modeling of post-tensioning

In a typical WTT structure, the prestress loss is not equal at different segments along the height. This may cause a large deviation in the numerical results when the prestress is applied according to the traditional cooling method. In this study, the prestress is simulated using a modified subsection cooling method. The effective prestress force is first calculated according to the prestress loss

Table 2 Reinforcement details of WTT-H160/C80

Tower section	Circumferential distributed rebar		Longitudinal distributed rebar	
	Diameter	Spacing [mm]	Diameter	Amount
1	D12	150	D12	100+100
2	D12	150	D12	90+90
3	D12	150	D12	90+90
4	D12	150	D12	90+90
5	D12	150	D12	80+80
6	D12	150	D12	80+80
7	D12	150	D12	80+80
8	D12	150	D12	80+80
9	D12	150	D12	80+80
10	D12	150	D12	80+80
11	D12	150	D12	80+80
12	D12	150	D12	80+80
13	D12	150	D12	70+70
14	D12	150	D12	70+70
15	D12	150	D12	70+70
16	D12	150	D12	70+70
17	D12	150	D12	70+70
18	D12	150	D12	70+70
19	D12	150	D12	70+70
20	D12	150	D12	70+70
21	D12	150	D12	60+60
22	D12	150	D12	60+60
23	D12	150	D12	60+60
24	D12	150	D12	110
25	D12	150	D12	110
26	D12	150	D12	110
27	D12	150	D12	110
28	D12	150	D12	110

theory. Next, the prestressing tendons are divided into 4-m sections based on the reference plane in the ABAQUS model, at which point the effective prestress force of each section can be obtained. Therefore, the corresponding cooling temperature is obtained. The cooling temperatures of the prestressing tendons in WTT-H160/C80 and WTT-H160/UHPC are listed in Table 3.

2.2 Results and discussion

Static analysis was performed on both FE models of WTT-H160/C80 and WTT-H160/UHPC. Here, the stresses of three typical sections of WTT, which were the bottom section with a door opening (Height: 0~4.0 m), a typical section at the middle of WTT (Height: 4.0~8.0 m), and the section at the top of WTT (Height: 104~110.5m), are discussed. Through a numerical simulation, the principal tensile stress, principal compressive stress, and horizontal displacement of WTT-H160/C80 and WTT-H160/UHPC

Table 3 Temperature reduction of prestressing tendons

Tower section	$\sigma_{p0,1}$ [MPa]	$\sigma_{p0,2}$ [MPa]	WTT-H160/C80	WTT-H160/UHPC
			cooling [°C]	cooling [°C]
1	1056.51	1084.84	-452	-464
2	1044.66	1075.42	-446	-460
3	1036.86	1067.74	-443	-456
4	1029.07	1060.06	-440	-453
5	1020.65	1051.96	-436	-450
6	1012.22	1043.86	-433	-446
7	1004.41	1036.19	-429	-443
8	996.60	1028.52	-426	-440
9	988.79	1020.87	-423	-436
10	980.98	1013.22	-419	-433
11	973.17	1005.58	-416	-430
12	965.37	997.95	-413	-426
13	956.80	989.80	-409	-423
14	948.22	981.65	-405	-420
15	940.39	974.03	-402	-416
16	932.56	966.42	-399	-413
17	924.74	958.83	-395	-410
18	916.92	951.25	-392	-407
19	909.11	943.69	-389	-403
20	901.31	936.15	-385	-400
21	892.50	927.95	-381	-397
22	883.68	919.74	-378	-393
23	875.83	912.23	-374	-390
24	867.42	904.35	-371	-386
25	859.00	896.49	-367	-383
26	851.14	889.05	-364	-380
27	843.29	881.64	-360	-377
28	845.24	877.72	-361	-375

* Note: $\sigma_{p0,1}$ and $\sigma_{p0,2}$ are the decompression prestress values of the prestressing tendons in the WTT-H160/C80 and WTT-H160/UHPC, respectively.

were obtained.

(1) Principal tensile stress results

The principal tensile stresses of the bottom, middle, and top sections of WTT with different concrete materials are respectively shown in Figs. 5-7. The principal tensile stress of WTT-H160/UHPC is around 0.50 MPa greater than that of WTT-H160/C80. The tensile strength of UHPC is much greater than that of C80 concrete, which is attributable to the addition of steel fibers in UHPC. In general, the tensile strength of UHPC is much larger than that of high strength concrete, with a difference exceeding 0.5 MPa. From this point of view, UHPC could replace the concrete and reinforcing in a WTT structure.

(2) Principal compressive stress results

Figs. 8-10 present the principal compressive stress results of the different sections of a WTT. The results

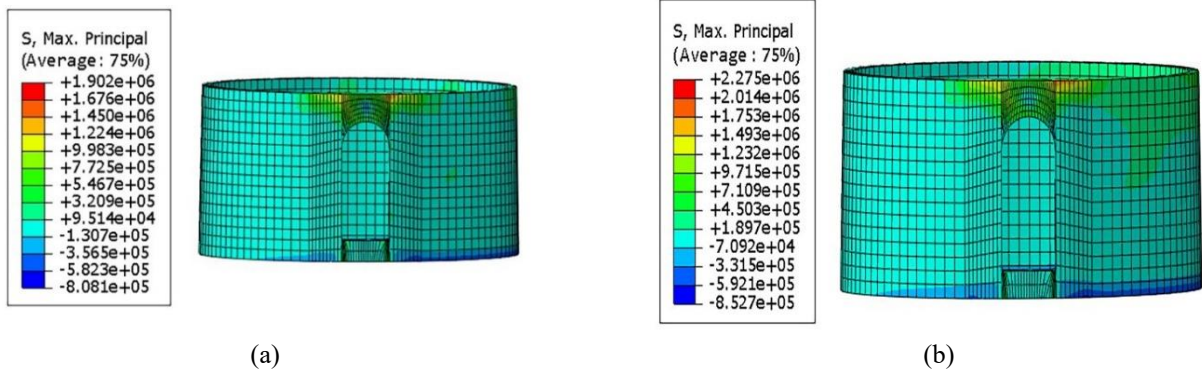


Fig. 5 Principal tensile stress of bottom section at 0~4.0 m: (a) WTT-H160/C80; and (b) WTT-H160/UHPC

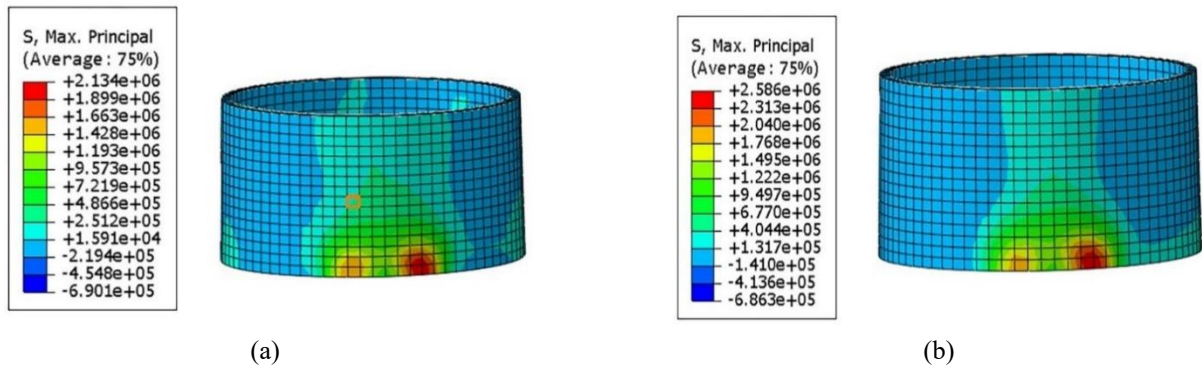


Fig. 6 Principal tensile stress of the typical section at the middle of WTT (4.0~8.0 m): (a) WTT-H160/C80; and (b) WTT-H160/UHPC

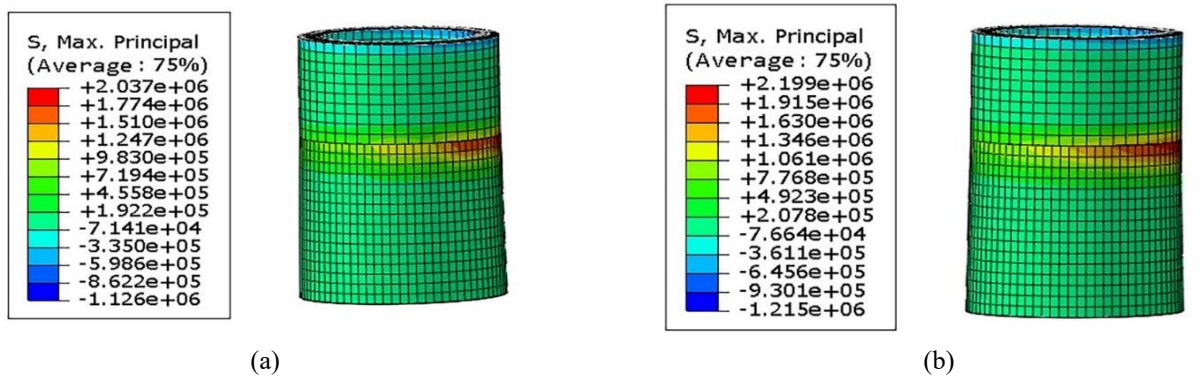


Fig. 7 Principal tensile stress at the top section of WTT (104.0~110.5 m): (a) WTT-H160/C80; and (b) WTT-H160/UHPC

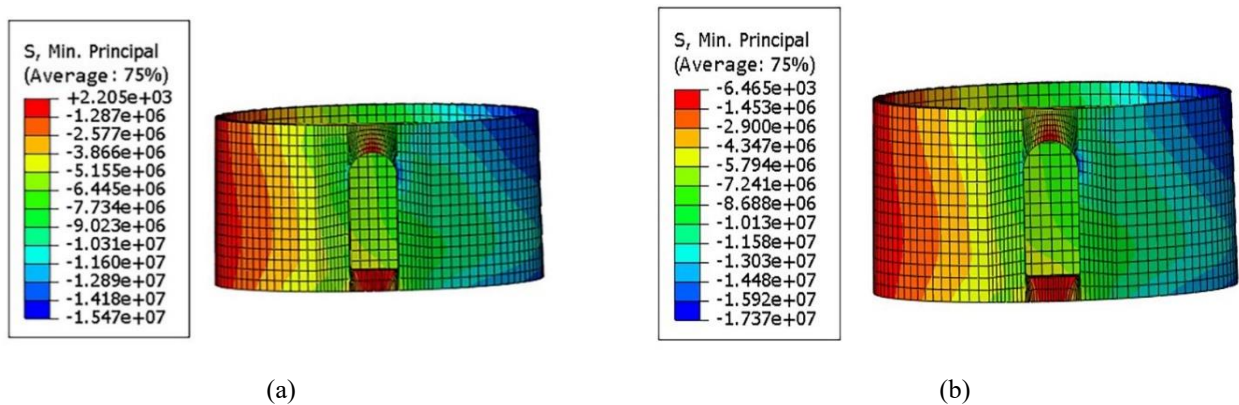


Fig. 8 Comparison of principal compressive stresses in the tower section of 0-4m: (a) WTT-H160/C80; and (b) WTT-H160/UHPC

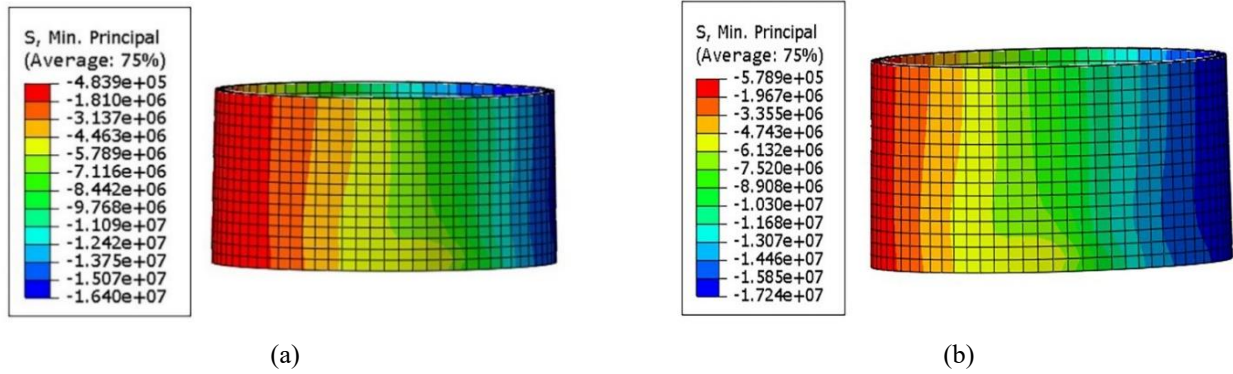


Fig. 9 Comparison of principal compressive stresses in the tower section of 4-8m: (a) WTT-H160/C80; and (b) WTT-H160/UHPC

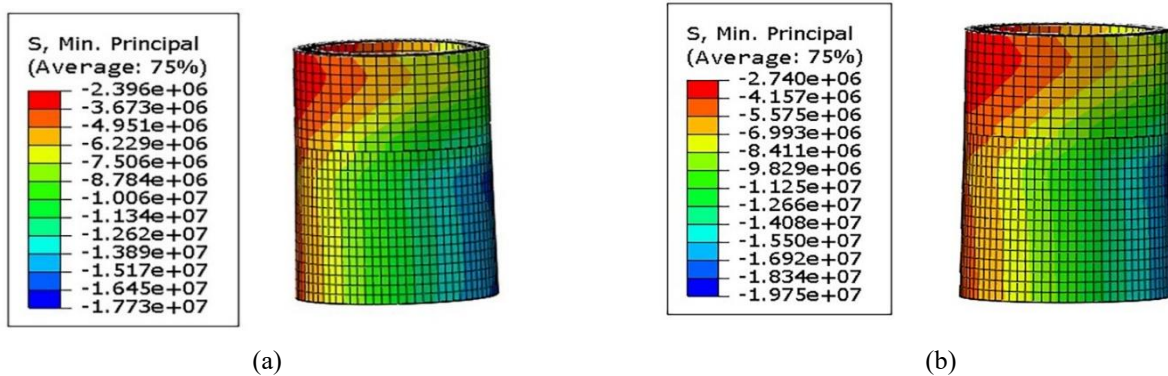


Fig. 10 Comparison of principal compressive stresses in the tower section of 104-110.5m: (a) WTT-H160/C80; and (b) WTT-H160/UHPC

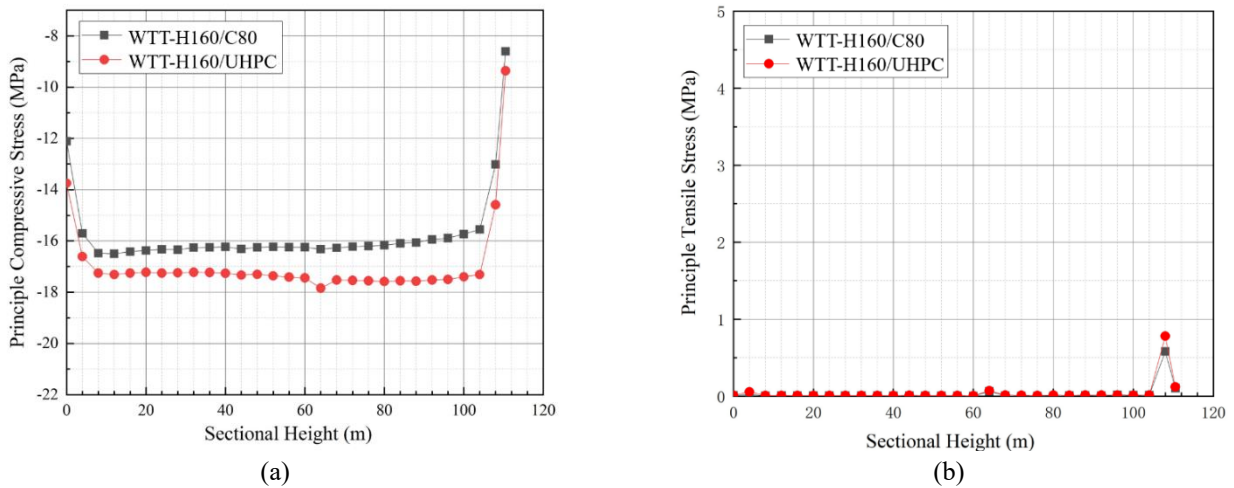


Fig. 11 Comparison of principal stress results: (a) principal compressive stress; and (b) principal tensile stress

indicate that the principal compressive stress of WTT-H160/UHPC was 1.0-2.0 MPa larger than that of WTT-H160/C80 concrete. The maximum principal compressive stress of WTT-H160/UHPC is less than 20 MPa under normal operating conditions, satisfying the requirements for serving conditions.

(3) Principal stress

Fig. 11 plots the principal tensile and compressive stresses at different horizontal joints along the height of a WTT. The principal compressive stresses of WTT do not

change significantly along the height except at the top and bottom segments, at which a sharp increase and a sharp drop respectively occur. At the same height, the principal compressive stress of WTT-H160/UHPC is 1.0~2.0 MPa larger than that of WTT-H160/C80. It can be seen from Fig. 11(b) that the principal tensile stresses of both types of WTTs remain constant along the height direction below 104 m. Further, a sudden fluctuation can be seen in both WTTs at 104~108 m. This might be caused by the change in the section thickness leading to a sudden change in the local

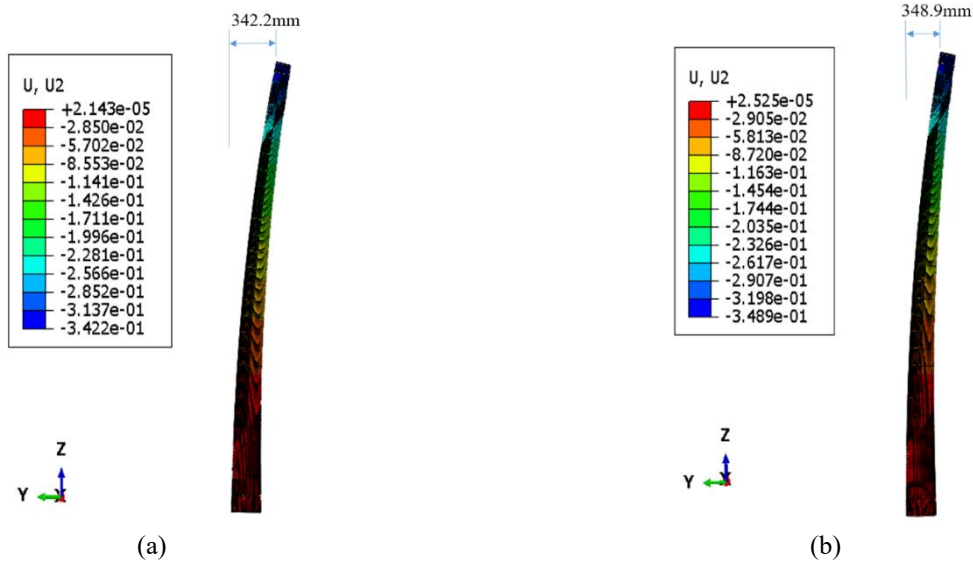


Fig. 12 Comparison of horizontal displacement results of concrete section of tower: (a) WTT-H160/C80; and (b) WTT-H160/UHPC

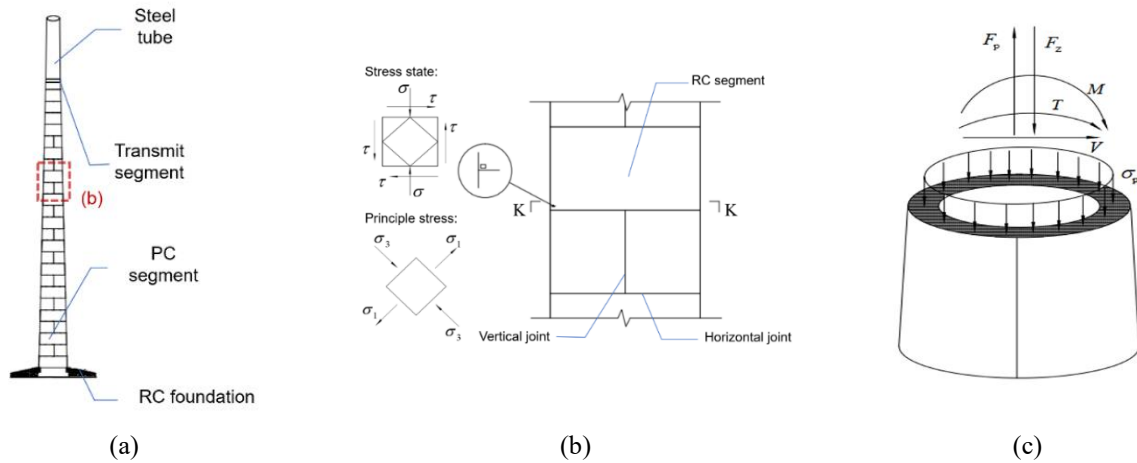


Fig. 13 Schematic diagram of a WTT tower: (a) typical WTT structure; (b) stress of unit-body near the joint; and (c) load at the cross-section of a horizontal joint

stress. At the same height, the principal tensile stress of WTT-H160/UHPC is only slightly large than that of WTT-H160/C80.

(4) Horizontal displacement

The horizontal displacements of the two different types of WTTs are shown in Fig. 12. The figure shows that the horizontal displacement of WTT-H160/UHPC is only slightly larger than that of WTT-H160/C80 at the same height. The maximum horizontal displacements of WTT-H160/C80 and WTT-H160/UHPC are 342.2 mm and 348.9 mm, respectively.

3. Design approach

3.1 Normal stress in horizontal joint section

To calculate the normal stress in the horizontal joint of a

WTT, a unit body was taken at the cross-section of horizontal joints. Fig. 13(b) shows the stress state and cross-section stress in the vicinity of the horizontal joints and Fig. 13(c) shows the load at the horizontal joint. Here, F_p is the vertical resultant force of prestressing tendons; F_z is the vertical concentrated force of the tower; M is the bending moment of the tower section; V is the shear force of the tower section; T is the torque of the tower section; σ_{pc} is the normal stress generated on the concrete section due to the tension of the prestressing tendons; and σ_{Mz} is the normal stress produced by the bending moment in the concrete section.

(1) Normal stress of cross-section

Under normal operating conditions, the weight and wind turbine generate bending moment M and vertical force F_z at the cross-section, which contribute to the normal stress. The prestress load also generates normal stress at the cross-section of a WTT. Therefore, the normal stress on the

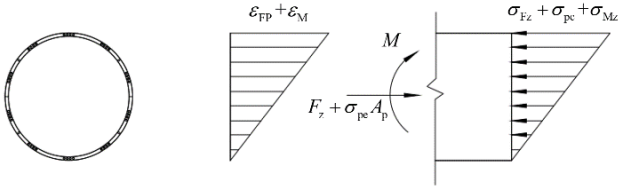


Fig. 14 Horizontal cross-section force diagram of a tower tube during the cross-section pressure reduction

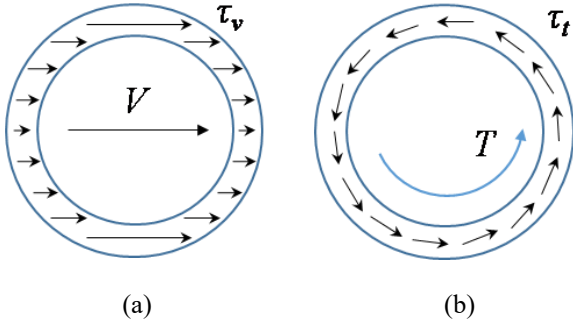


Fig. 15 Shear stress caused by shear and torsion: (a) cross-sectional shear stress flow under shear force; and (b) cross-sectional shear stress flow under torque

tension and compression sides of the section can be expressed as:

$$\begin{aligned}\sigma_{z,t} &= \sigma_{Mz} - \sigma_{pc} - \sigma_{Fz} \\ \sigma_{z,c} &= -\sigma_{Mz} - \sigma_{pc} - \sigma_{Fz}\end{aligned}\quad (1)$$

where σ_{Fz} is the normal stress caused by the vertical concentrated load and is equal to F_z/A_0 , where F_z is the vertical concentrated load and A_0 is the converted area of the section; σ_{Mz} is the normal stress generated by the bending moment and is equal to My/I_0 , where y is the distance from the outermost edge of the ring section to the neutral axis and I_0 is the converted moment of inertia of the section calculated by equating the steel bar to concrete; and σ_{pc} is the normal stress on the section generated by the prestressing tendon on the section and is equal to $\sigma_{pe}A_p/A_N$, where A_p is the cross-sectional area of prestressing tendons, A_N is the net cross-sectional area of the section, for which the area of prestressing tendons is deducted from the total area; and σ_{pe} is an effective prestress.

(2) Cracking moment M_{cr} at the cross-section

As can be seen in Fig. 13(c), at the horizontal section, only the bending moment M , the vertical concentrated load F_z , and the action of prestressed tendons will generate the normal stress σ_{pc} . When $\sigma_{Mz} = \sigma_{Fz} + \sigma_{pc}$, the tensile stress at the edge of the section is zero, as shown in Fig. 14.

Therefore, the cracking moment of the joint section can be obtained as:

$$M_{cr} = (\sigma_{Fz} + \sigma_{pc})W_0 \quad (2)$$

where W_0 is the equivalent modulus of the calculated section (steel bar being equivalent to the concrete area).

3.2 Shear stress of horizontal joint section

The conventional tower tube (wall thickness=220 mm and outer diameter=8 m) can be approximately considered as a thin-walled ring. The shear stress generated by shear and torsion is illustrated in Fig. 15.

The maximum shear stress of the section under shear force can be expressed as:

$$\tau_{v,max} = \frac{VS_z^*}{2I_z t} \quad (3)$$

Where

$$S_z^* = \frac{2}{3} \left(R + \frac{t}{2} \right)^3 - \frac{2}{3} \left(R - \frac{t}{2} \right)^3 = 2R^2 t + \frac{t^3}{6} \approx 2R^2 t \quad (4)$$

$$I_z = \frac{\pi}{4} \left(R + \frac{t}{2} \right)^4 - \frac{\pi}{4} \left(R - \frac{t}{2} \right)^4 = \pi R^3 t + \frac{\pi}{4} R t^3 \approx \pi R^3 t \quad (5)$$

where t and R are the thickness and radius of thin-walled ring, respectively; S_z^* is the first moment of the section; and I_z is the second moment of the section.

By substituting Eqs. (4) and (5) into Eq. (3), the maximum shear stress of the section under shear force can be obtained as:

$$\tau_{v,max} = \frac{2V}{A} \quad (6)$$

where A is the area of the section of the shear member and is equal to $2\pi R t$.

As shown in Fig. 15(b), the shear stresses caused by torque on the inner and outer sides of the ring are not equal. Due to the small thickness of the tower wall, it is considered that the shear stress distribution on the inner and outer sides is considered to be approximately equal. Therefore, the shear stress caused by torque is as follows:

$$\tau_T = \frac{T}{W_t} \quad (7)$$

where W_t is the moment of plastic resistance of the section of the torsion member.

The maximum shear stress of the corresponding section under the combined action of shear and torsion is:

$$\tau_{VT,max} = \tau_{v,max} + \tau_T = \frac{2V}{A} + \frac{T}{W_t} \quad (8)$$

For conservatism, the maximum shear stress $\tau_{v,max}$ is approximately used as the calculated shear stress of the cross-section in the principal stress checking calculation. According to the "Specification for Design of Concrete Structures" (GB50010-2010), the cross-sectional stress of prestressed concrete tower tube under normal operating conditions should meet the following conditions:

$$\sigma_{tp} \leq \beta_t f_{tk}, \quad \sigma_{cp} \leq \beta_c f_{ck} \quad (9)$$

where σ_{tp} and σ_{cp} are the principal tensile stress and principal compressive stress of concrete, respectively; β_t is the reduction coefficient of the torsion load-bearing capacity of concrete; β_c is the influence coefficient of concrete strength; and f_{tk} and f_{ck} are the standard values

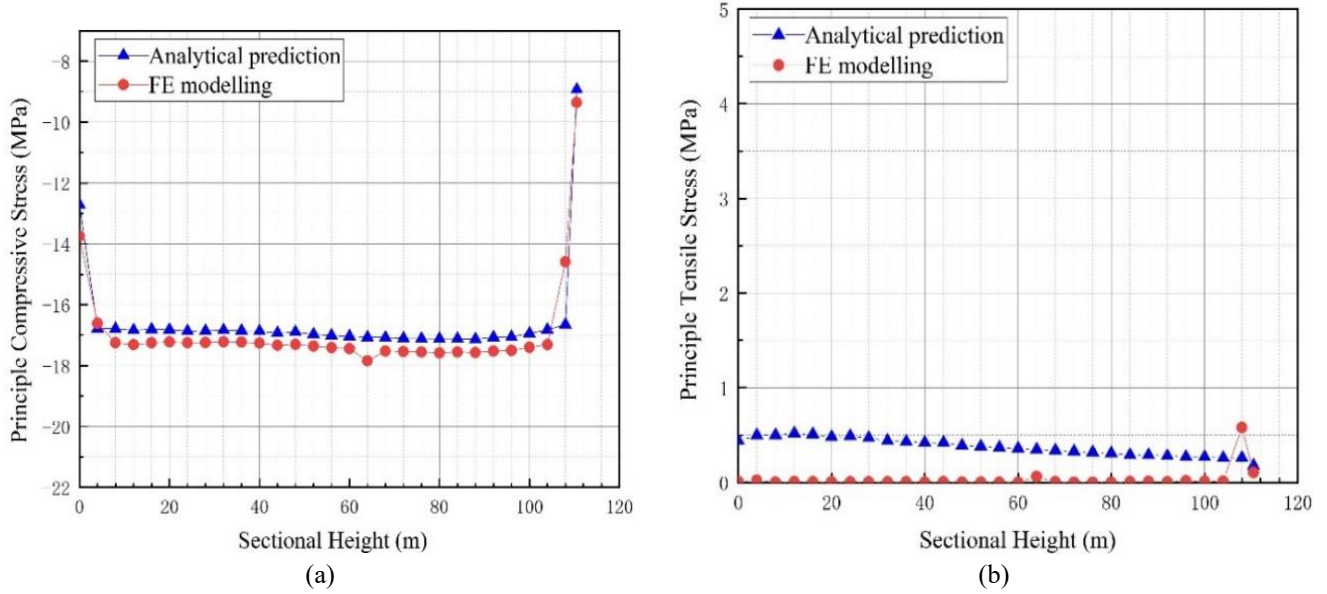


Fig. 16 Comparison of principal stresses between the analytical model and the FE model of WTT-H160/C80: (a) principal compressive stress; and (b) principal tensile stress

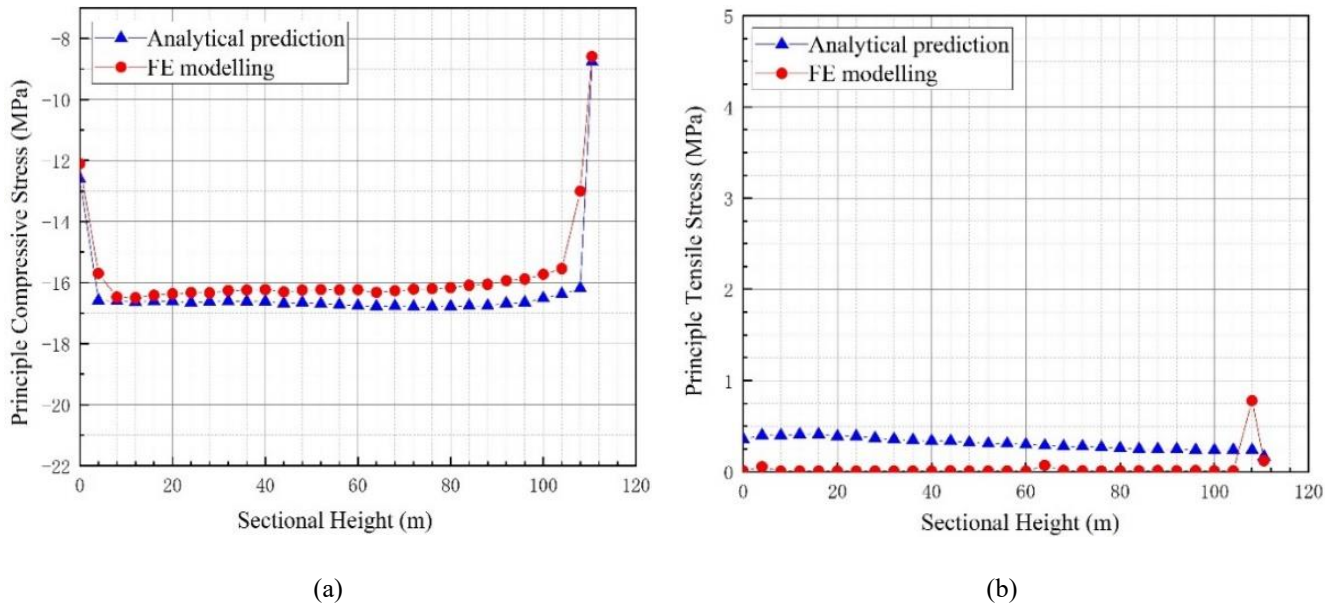


Fig. 17 Comparison of principal stresses between the analytical model and the FE model of WTT-H160/UHPC: (a) principal compressive stress and (b) principal tensile stress

of the tensile strength and compressive strength of concrete, respectively.

Under the action of the combined bending moment, the control stress on the tensile side of the tower is the main tensile stress whereas the control stress on the compressive side of the tower is the main compressive stress. They can be respectively calculated by:

$$\left. \begin{aligned} \sigma_{tp,t} \\ \sigma_{cp,t} \end{aligned} \right\} = \frac{1}{2} \left(\frac{My}{I_0} - \frac{F_z}{A_0} - \sigma_{pc} \right) \pm \frac{1}{2} \sqrt{\left(\frac{My}{I_0} - \frac{F_z}{A_0} - \sigma_{pc} \right)^2 + 4 \left(\frac{2V}{A} + \frac{T}{W_t} \right)^2} \quad (10)$$

$$\left. \begin{aligned} \sigma_{tp,c} \\ \sigma_{cp,c} \end{aligned} \right\} = \frac{1}{2} \left(-\frac{My}{I_0} - \frac{F_z}{A_0} - \sigma_{pc} \right) \pm \frac{1}{2} \sqrt{\left(-\frac{My}{I_0} - \frac{F_z}{A_0} - \sigma_{pc} \right)^2 + 4 \left(\frac{2V}{A} + \frac{T}{W_t} \right)^2} \quad (11)$$

where $\sigma_{tp,t}$ and $\sigma_{cp,t}$ are the principal tensile stress and principal compressive stress on the tensile side of the tower, respectively; and $\sigma_{tp,c}$ and $\sigma_{cp,c}$ are the principal tensile stress and principal compressive stress on the compression side, respectively.

3.3 Comparison of results between design approach and FE model

Figs. 16 and 17 show a comparison between the results

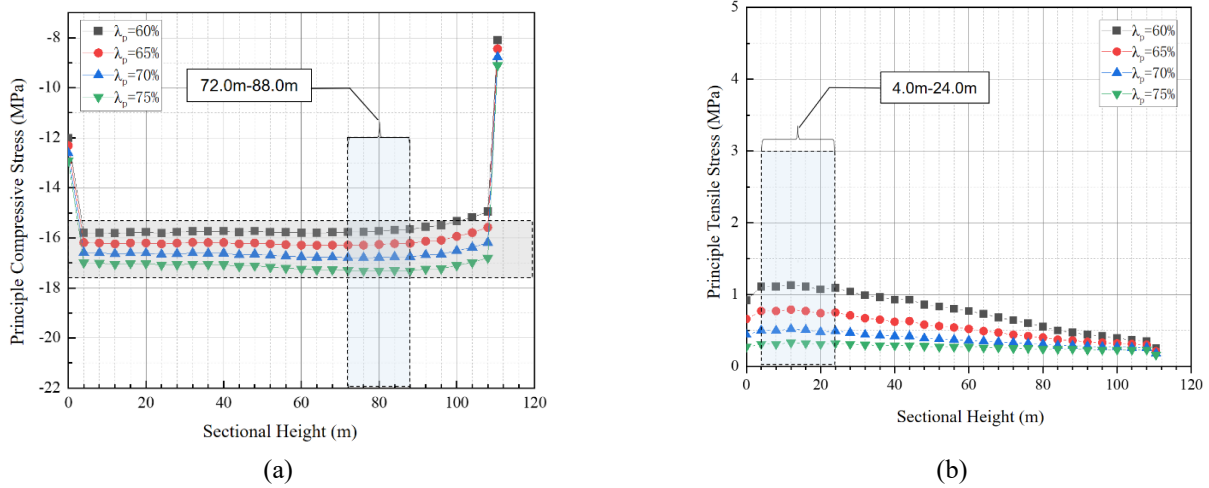


Fig. 18 (a) Principal compressive stress and (b) principal tensile stress under different degrees of prestress

of the proposed design approach and those from the FE modeling. As can be seen in the figures, for both types of WTT structures, the results of the proposed approach and FE modeling are consistent with each other. For the principal compressive strength, although the results of the proposed design approach slightly differ from those from the two FE models, the overall difference between the results of the two models is within 0.5 MPa.

On the other hand, for both types of WTT structures, the principal tensile stress obtained by the proposed design approach is slightly larger than that obtained by the FE model, with a difference of less than 0.5 MPa. However, it should be noted that the sudden change in tensile stress at 104~108 m height in the FE model could not be obtained by analytical calculations. Nevertheless, the results of the proposed design approach and FE model of WTTs are close to each other. This indicates that it is therefore feasible to use the design approach proposed in Section 3 to determine the stress of a WTT structure.

4. Effects of different factors on principal stress

4.1 Effect of effective prestress

The effective stress of the prestressing tendon is expressed as $\lambda_p f_{ptk}$, where f_{ptk} is the ultimate tensile strength of the tendon and λ_p is the ratio of effective to ultimate tensile stresses. To investigate the effect of λ_p , the principal compressive and tensile stresses with different values of λ_p (60%, 65%, 70%, and 75%) are calculated considering the prestress loss. The results are plotted in Fig. 18.

As can be seen in the figure, under the same degree of prestressing, the principal compressive stresses at 0 m and 110.5 m are both relatively small. When the prestressing tendons are post-tensioned, the prestress loss at both ends is the largest. Therefore, the effective stress of the prestressing tendon is small at both ends, ultimately resulting in relatively small compressive stresses. At other positions, with increasing height, the principal compressive stress

increases slightly below 72 m and decreases slightly above 88 m. It is also found that under different degrees of prestressing, the maximum principal compressive stress along the vertical direction of tower is in the range from 72~88 m.

On the other hand, the principal tensile stress decreases gradually with an increasing degree of prestressing. However, the decreasing rate is reduced when the degree of prestressing increases. The maximum principal tensile stress occurs in the height range from 4~24 m.

4.2 Effect of bending moment-to-shear ratio

The actual stress state of the WTT tubular structure is a combined stress state under the action of bending moment and shear force. This section discusses the effects of the bending moment-to-shear ratio $\lambda(h)$ of the section on the principal stress of the cross-section of horizontal joints. The bending moment-to-shear ratio $\lambda(h)$ can be calculated by:

$$\lambda(h) = \frac{M(h)}{V(h)h} \quad (12)$$

where $M(h)$ and $V(h)$ are the bending moment value and shear force value of the section at the height of h , respectively.

The bending moment ratio is affected by the bending moment and the shear force. By changing the value of the bending moment while keeping the shear force constant, four different bending moment-to-shear ratios— $0.8\lambda(h)$, $1.0\lambda(h)$, $1.2\lambda(h)$, and, $1.5\lambda(h)$ —are first assumed. For example, the bending moment is increased by 50% for the bending moment-to-shear ratio of $1.5\lambda(h)$. The principal stresses along the height of WTT are calculated under four different bending moment-to-shear ratios, as shown in Fig. 19. As seen, at each bending moment-to-shear ratio, the principal compressive stress almost holds constant along the height between 4 m and 96 m; however, it reduces significantly at the top and bottom of the tower. The principal tensile stress increases at the bottom of the tower and then decreases gradually with increasing height. At the

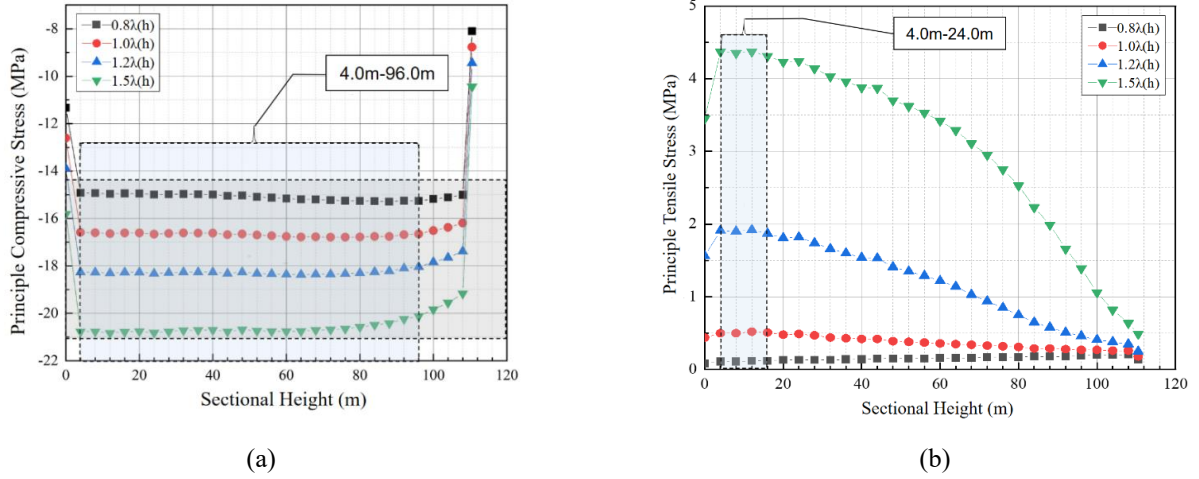


Fig. 19 (a) Principal compressive stress and (b) principal tensile stress under different bending moment-to-shear ratios (changed by bending moments)

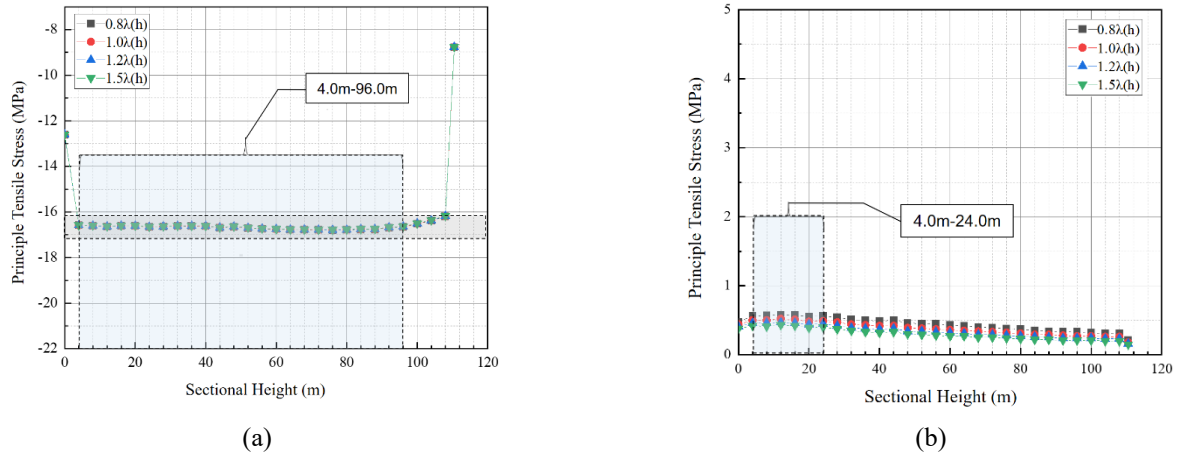


Fig. 20 (a) Principal compressive stress and (b) principal tensile stress under different bending moment-to-shear ratios (changed by shear forces)

same height, both the principal compressive and tensile stresses increase significantly with an increase in bending moment-to-shear ratio. Their increasing rates also increase when the bending moment-to-shear ratio increases.

When changing the shear force, the bending moment remains constant. For example, the shear force is decreased by 1/3 for the bending moment-to-shear ratio of $1.5\lambda(h)$. The principal stresses of the sections along the height of WTT under four different bending moment-to-shear ratios are shown in Fig. 20. As can be seen in the figure, the change in shear force does not affect the principal compressive stress, but it slightly affects the principal tensile stress. The principal tensile stress is slightly reduced with the increasing bending moment-to-shear ratios caused by the reduction of shear forces. It is also found that the principal tensile stresses are relatively small in this case. By comparing Figs. 19 and 20, it can be seen that regarding the effect of bending moment-to-shear ratios, the change in bending moments shows a more significant effect on the principal stress than the change of shear forces. As is evident, the principal stress is controlled by bending moments.

4.3 Effect of shear-to-torsion ratio

In this study, the shear-to-torsion ratio $\zeta(h)$ is defined as:

$$\zeta(h) = \frac{V(h)h}{T(h)} \quad (13)$$

where $T(h)$ is the torque of the section at the height of h .

The shear-to-torsion ratio is affected by both shear force and torsion. In this section, four different shear-to-torsion ratios— $0.5\zeta(h)$, $1.0\zeta(h)$, $2.0\zeta(h)$, and $4.0\zeta(h)$ —are considered by either increasing the shear force or reducing the torsion.

Fig. 21 shows the principal stresses along the height under different shear-to-torsion ratios with only the shear forces changing. As can be seen, the change in shear-to-torsion ratios with the change in shear forces does not affect the principal compressive stress. The principal compressive stress almost remains constant when the shear-to-torsion ratio changes. However, the principal tensile stress increases significantly with an increasing shear-to-torsion

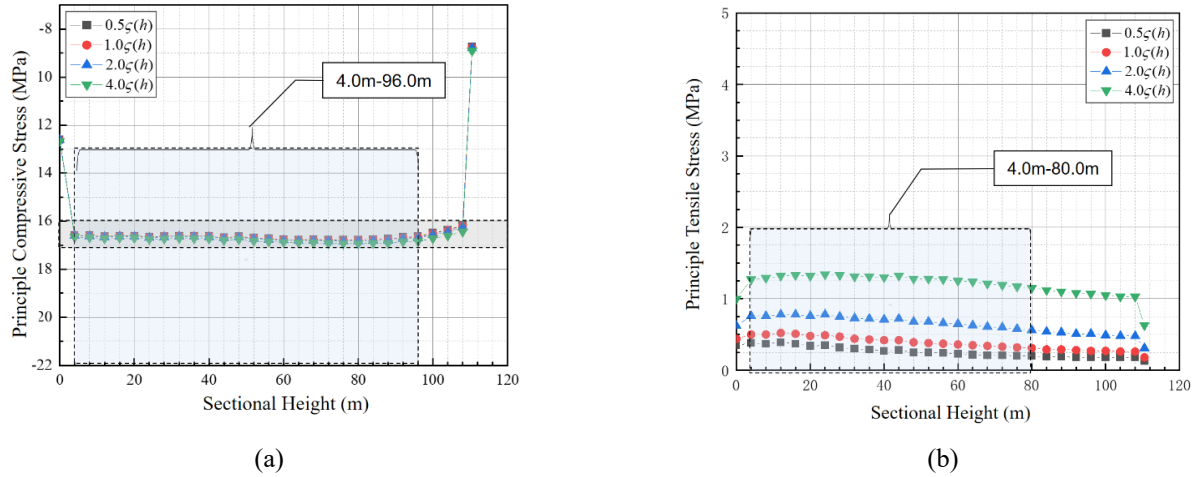


Fig. 21 (a) Principal compressive stress; and (b) principal tensile stress under different shear-to-torsion ratios (changed by shear forces)

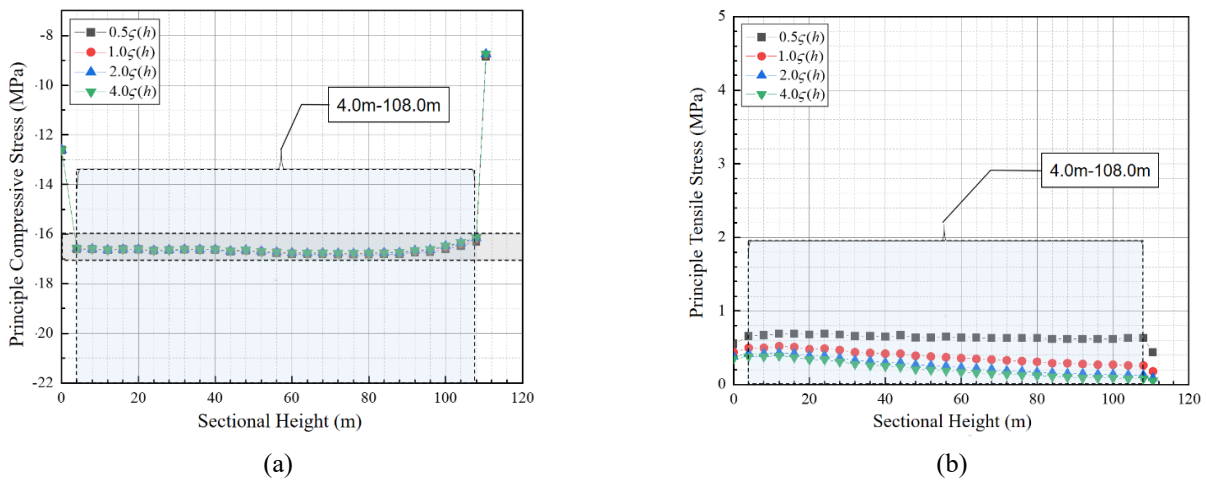


Fig. 22 (a) Principal compressive stress; and (b) principal tensile stress under different shear-to-torsion ratios (changed by torsions)

ratio under this condition.

Fig. 22 shows the principal stresses along the height under different shear-to-torsion ratios caused by the change in torque. Similar to the case under different shear forces, the principal compressive stress does not change with different shear-to-torsion ratios with changing torque. However, the principal tensile stress decreases gradually when the shear-to-torsion ratio increases.

According to Figs. 21 and 22, it can be seen that the change in shear-to-torsion ratios has little influence on the principal compressive stress. This is irrespective of the change in shear forces or torques. However, the shear-to-torsion ratio has a relatively significant influence on the principal tensile stress. The principal tensile stress is more sensitive to the changes of shear force than the changes in torques.

5. Conclusions

This study focused on the design approach of hybrid

WTT structures. By conducting FE modeling and analytical analysis, the following conclusions have been drawn:

(1) Under the same external load and design conditions of WTT geometry, the principal tensile stress, principal compressive stress, and horizontal displacement of WTT-H160/UHPC are slightly larger than those of WTT-H160/C80. The difference in principal tensile stress is within 0.5 MPa and the difference in principal compressive stress is within 1.0~2.0 MPa. UHPC can be used to replace the concrete and reinforcing bars in a WTT while achieving similar performance.

(2) A design approach for the calculation of stress of the horizontal joints of a WTT is proposed. The results of the proposed design approach and FE model are consistent for both WTT-H160/C80 and WTT-H160/UHPC.

(3) The degree of prestressing affects both principal compressive stress and principal tensile stress. With an increase in the degree of prestressing, the principal compressive stress gradually increases while the principal tensile stress decreases.

(4) The principal stress of the WTT is more sensitive to

the bending moment of the section. Increasing the bending moment, which leads to an increase in the bending moment-to-shear ratio, increases both principal compressive stress and principal tensile stress. However, changing the shear force does not significantly affect the principal stress.

(5) The change in shear-to-torsion ratio mainly affects the principal tensile stress of a WTT but does not significantly affect the principal compressive strength. With an increase in the shear force, the corresponding increase in the shear-to-torsion ratio increases the principal tensile stress, while with a decrease in torque, the corresponding increase of shear-to-torsion ratio gradually decreases the principal tensile stress.

(6) In this study, two FE models are analyzed under normal operating conditions where all tower materials are assumed to be elastic. The results are consistent with the proposed design approach. However, to verify the feasibility of using UHPC in the WTT, it is necessary to consider the behavior of the tower in the inelastic range.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Notations

		$\tau_{v,max}$	maximum shear stress of the section under shear force
		$\tau_{vT,max}$	maximum shear stress of the corresponding section under the combined action of shear and torsion
A	area of the section of the shear member		
A_0	converted area of the section		
A_N	net cross-sectional area of the section		
A_p	cross-sectional area of prestressing tendons		
f_{ck}	standard values of the compressive strength of concrete		
f_{tk}	standard values of the tensile strength of concrete		
F_p	vertical resultant force of prestressing tendons		
F_z	vertical concentrated force of the tower		
h	height		
I_0	converted moment of inertia of the section calculated by equating the steel bar to concrete		
I_z	second moment of the section		
M	bending moment of the tower section		
M_{cr}	cracking moment		
$M(h)$	bending moment value of the section at the height of h		
R	radius of thin-walled ring		
S_z^*	first moment of the section		
t	thickness of thin-walled ring		
T	torque of the tower section		
$T(h)$	torque of the section at the height of h		
V	shear force of the tower section		
$V(h)$	shear force value of the section at the height of h		
W_0	equivalent modulus of the calculated section		
W_t	moment of plastic resistance of the section of the torsion member		
y	distance from the outermost edge of the ring section to the neutral axis		
β_c	influence coefficient of concrete strength		
β_t	reduction coefficient of the torsion load-bearing capacity of concrete		
λ_p	ratio of effective to ultimate tensile stresses		
$\lambda(h)$	bending moment-to-shear ratio		
σ_{Fz}	normal stress caused by the vertical concentrated load		
σ_{Mz}	normal stress produced by the bending moment in the concrete section		
σ_{cp}	principal compressive stress of concrete		
σ_{tp}	principal tensile stress of concrete		
$\sigma_{cp,c}$	principal compressive stress on the compression side of the tower		
$\sigma_{tp,c}$	principal tensile stress on the compression side of the tower		
$\sigma_{cp,t}$	principal compressive stress on the tensile side of the tower		
$\sigma_{tp,t}$	principal tensile stress on the tensile side of the tower		
σ_{pc}	normal stress generated on the concrete section due to the tension of the prestressing tendons		
σ_{pe}	effective prestress		
$\zeta(h)$	shear-to-torsion ratio		
τ_T	shear stress caused by torque		

